Chapter 4: 8, 16, 18, 26

8. Input

16. The advantage of the bit addressing is that it allows each bit to be modified without affecting the other bits.

18.

```
SBI DDRB, 2
SBI DDRB, 5
L1: SBI PORTB, 2
SBI PORTB, 5
CBI PORTB, 2
CBI PORTB, 5
RJMP L1
```

26.

```
; SOLUTION TO 4.26
CBI DDRB, 5
CBI DDRB, 6
SBI PORTB, 5
SBI PORTB, 6
LDI R16, 0xFF
OUT DDRC, R16
L1:  SBIC PINB, 5
RJMP L2; JUMP L2 IF PB5 HIGH
IS_LOW: LDI R16, 0x55
OUT PORTC, R16
RJMP L1
L2: SBIS PINB, 6
RJMP IS_LOW; JUMP IS_LOW IF PB6 IS LOW
LDI R16, 0xAA
OUT PORTC, R16
RJMP L1
```

(g) 0011 0111 OR 0010 0110 = 0011 0111 = 0x37

22. both (a) and (b)

24.

(a) 0 01010110 □ 0 01100101 □ C = 0 □ 1 00110010 □ 0 10011001 = 99h

Chapter 5: 19 a,c,e; 22 b; 24 a

19 a,c,e.

(a) 0110 0101 AND 0111 0110 = 0110 0100 = 0x64
(b) 0111 0000 OR 0110 1011 = 0111 1011 = 0x7B
(c) 1001 0101 EXOR 1010 1010 = 0011 1111 = 0x3F
(d) 0101 1101 AND 0111 0101 = 0101 0101 = 0x55
(e) 1100 0101 OR 0001 0010 = 1101 0111 = 0xD7
(f) 0110 1010 EXOR 0110 1110 = 0000 0100 = 0x04

22. both (a) and (b)

24.
(a) 0 01010110 □ 0 01100101 □ C = 0 □ 1 00110010 □ 0 10011001 = 99h

Additional problems:

1. Explain the role of DDRx, PORTx, and PINx in I/O operations.

   DDRx – used to configure the pins for PORTx as input or output (0=>input, 1=>output). Can configure individual pins or all of the pins for PORTx.

   PORTx – used as both an internal AVR register for storing values (PORTx is configured as an input port) as well as an output port (outputting the PORTx value to the port pins). Also, used to enable/disable the pull-up resistors for input port pins. Bit-wise or word-wise port bit configuration can be performed.

   PINx – used to received data values from the port pins when the port is configured as an input port. Port pins may be configured to use or not use pull-up resistors for stabilizing the received data values. Bit-wise or word-wise port bit configuration can be performed.

2. A switch is connected to pin PC4 and an LED to pin PC5. Write a program to get the status of the switch (SW) and send it to the LED. Assume that the LED is common anode (0 turns the LED on and 1 turns the LED off). When SW = 1, the LED should be on. When SW = 0, the LED should be off.

```
SBI DDRC, 5; PC5 output
CBI DDRC, 4; PC4 input
SBI PORTC, 4; pull-up resistor for PC4 enabled

LOOP:   SBIS PINC,4; SEE IF SWITCH IS SET
        RJMP TURNLEDOFF
        CBI PORTC,5; TURNLEDON- ACTIVE LOW LED
        RJMP LOOP
TURNLEDOFF: SBI PORT,5; TURNLEDOFF-ACTIVE LOW LED
        RJMP LOOP
```

3. Work with your project team to create an Microchip Studio workspace for the ATmega32. Copy the code from your solution to Chapter 4: Problem 26. Single Step execute the program and use the Microchip Studio simulation for the I/O PINS to change the input pin values and observe the output PORT bit values. Submit a screen shot of your program simulation showing the PIN and PORT registers.

4. Which of the following are invalid uses of the single register/immediate addressing mode?
   a) LDI R16, 0x32 b) STS 0x62, R17 c) STS 0x42, 0x01 d) OUT 0x15, R19

   e) OUT PORTB, 0x35 f) OUT 0x18, 0x19

   c-0x01 must be a GPR (single register)

   e-0x35 must be a GPR (single register)

   f-0x19 must be a GPR (single register)

5. Identify the addressing mode for each of the following (single register/immediate; register, direct address, register indirect):

   a) OUT PORTA, R19 b) STS 0x20, R19 c) SUB R18, R31 d) MOV R1, R2

   e) ST Z, R19

   OUT PORTA, R19 – DIRECT (ALSO SINGLE REGISTER)

   STS 0X20, R19-DIRECT (ALSO SINGLE REGISTER)

SUB R18, R31-REGISTER-BOTH GPRs

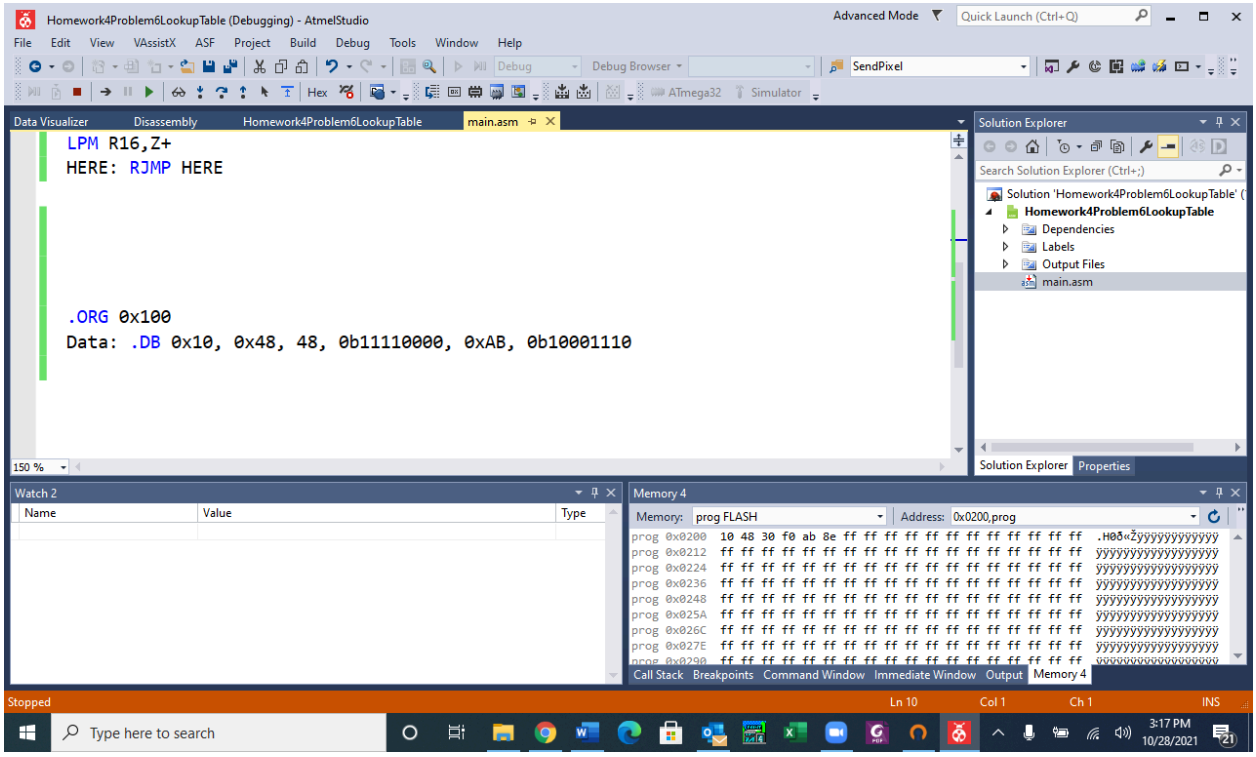
MOV R1, R2-REGISTER-BOTH GPRs

ST Z, R19-REGISTER INDIRECT (ALSO SINGLE REGISTER)

6. For the following assembler directives, determine the contents for each ROM location:

.ORG 0x100

Data: .DB 0x10, 0x48, 48, 0b11110000, 0xAB, 0b10001110



EACH CODE MEMORY LOCATION HAS A 16 BIT WORD ADDRESS (0X100 ABOVE) THAT HAS AN ASSOCIATED 8 BIT ADDRESS FOR THE LOW BYTE AND HIGH BYTE. THE LOW BYTE ADDRESS IS 2 TIMES THE 16 BIT ADDRESS. THE HIGH BYTE ADDRESS IS 2 TIMES THE 16 BIT ADDRESS + 1.

0x10 – LOCATED AT 0X200

0X48-LOCATED AT 0X201

48-LOCATED AT 0X202

0B11110000-LOCATED AT 0X203

0XAB-LOCATED AT 0X204

0B10001110-LOCATED AT 0X205

7. Which of the following instructions are valid? If the instruction is valid, which bit is altered?

   a) CBI PORTB, 5 b) CBR R16, 3 c) SBR R21, 0 d) BLD PORTA, 2

A-VALID – PB5

B-VALID-BIT 3, R16

C- VALIE BIT 0, R21

D-INVALID, MUST USE A GPR NOT AN I/O REGISTER

8. Write an assembly program that finds the number of 0s in an 8-bit data item.

```
.ORG 0
LDI R16, 0XFF
OUT DDRC, R16; CONFIGURE PORTC AS OUTPUT PORT
; ASSUME PORTB IS AN OUTPUT PORT
LDI R16, 0XF0; ASSUME PORTB = 0X0F IS WORD TO COUNT THE NUMBER OF 0S
OUT PORTB, R16
; CREATE COPIES OF PORTB TO COUNT # OF 0S
IN R19, PORTB; YES, THIS IS EXTRA WORK TO INITIALIZE AND GET COPIES OF PORTB


LDI R17, 0X00; INITIALIZE NUMBER OF 0S IN WORD TO 0
LDI R18, 8; NUMBER OF BITS IN PORTB WORD TO COUNT

LOOP:  SRBS R19,0
INC R17; INCREMENT # 0S
ROR R19
DEC R18; DECREMENT # BITS IN WORD
BRNE LOOP; COMPLETE FOR ALL 8 BITS OF WORD TO COUNT 0S
OUT PORTC, R17; OUTPUT #0S TO PORTC
```

SECTION 4.1: I/O PORT PROGRAMMING IN AVR

1. The ATmega32 has a DIP package of ____ pins.
2. In ATmega32, how many pins are assigned to V_{CC} and GND?
3. In the ATmega32, how many pins are designated as I/O port pins?
4. How many pins are designated as PORTA in the 40-pin DIP package and what are their numbers?
5. How many pins are designated as PORTB in the 40-pin DIP package and what are their numbers?
6. How many pins are designated as PORTC in the 40-pin DIP package and what are their numbers?
7. How many pins are designated as PORTD in the 40-pin DIP package and what are their numbers?
8. Upon reset, all the bits of ports are configured as _____ (input, output).
9. Explain the role of DDRx and PORTx in I/O operations.

10. Write a program to get 8-bit data from PORTC and send it to PORTB and PORTD.
11. Write a program to get 8-bit data from PORTD and send it to PORTB and PORTC.
12. Which pins are for RxD and TxD?
13. Give data memory location assigned to DDR registers of Ports A–C for the ATmega32.
14. Write a program to toggle all the bits of PORTB and PORTC continuously (a) using 0xAA and 0x55 (b) using the COM instruction.

SECTION 4.2: I/O BIT MANIPULATION PROGRAMMING

15. Which ports of the ATmega32 are bit-addressable?
16. What is the advantage of bit-addressability for AVR ports?
17. Is the instruction "`COM PORTB`" a valid instruction?
18. Write a program to toggle PB2 and PB5 continuously without disturbing the rest of the bits.
19. Write a program to toggle PD3, PD7, and PC5 continuously without disturbing the rest of the bits.
20. Write a program to monitor bit PC3. When it is HIGH, send 0x55 to PORTD.
21. Write a program to monitor the PB7 bit. When it is LOW, send $55 and $AA to PORTC continuously.
22. Write a program to monitor the PA0 bit. When it is HIGH, send $99 to PORTB. If it is LOW, send $66 to PORTB.
23. Write a program to monitor the PB5 bit. When it is HIGH, make a LOW-to-HIGH-to-LOW pulse on PB3.
24. Write a program to get the status of PC3 and put it on PC4.
25. Create a flowchart and write a program to get the statuses of PD6 and PD7 and put them on PC0 and PC7, respectively.
26. Write a program to monitor the PB5 and PB6 bits. When both of them are HIGH, send $AA to PORTC; otherwise, send $55 to PORTC.
27. Write a program to monitor the PB5 and PB6 bits. When either of them is HIGH, send $AA to PORTC; otherwise, send $55 to PORTC.
28. Referring to Figure 4-8 and Table 4-8, write the machine equivalent of "SBIS PINB,3".
29. Referring to Figure 4-6 and Table 4-8, write the machine equivalent of the "SBI PORTA,2" instruction.

SECTION 5.3: LOGIC AND COMPARE INSTRUCTIONS

19. Find the contents of register R20 after each of the following instructions:

(a)
```
LDI   R20,0x65
LDI   R21,0x76
AND   R20,R21
```

(b)
```
LDI   R20, 0x70
LDI   R21, 0x6B
OR    R20, R21
```

(c)
```
LDI   R20,0x95
LDI   R21,0xAA
EOR   R20,R21
```

(d)
```
LDI   R20, 0x5D
LDI   R21, 0x75
AND   R20, R21
```

(e)
```
LDI   R20,0x0C5
LDI   R21,0x12
OR    R20,R21
```

(f)
```
LDI   R20, 0x6A
LDI   R21, 0x6E
EOR   R20, R21
```

(g)
```
LDI   R20,0x37
LDI   R21,0x26
OR    R20,R21
```

20. Explain how the BRSH instruction works.
21. Does the compare instruction affect the flag bits of the status register?
22. Assume that R20 = 85H. Indicate whether the conditional branch is executed in each of the following cases:

(a)
```
LDI   R21, 0x90
CP    R20,R21
BRLO  NEXT
...
```

(b)
```
LDI   R21, 0x70
CP    R20, R21
BRSH  NEXT
...
```

23. For Problem 22, indicate the value in R20 after execution of each program.

SECTION 5.4: ROTATE AND SHIFT INSTRUCTIONS AND DATA SERIALIZATION

24. Find the contents of R20 after each of the following is executed:

(a)
```
LDI   R20, 0x56
SWAP  R20
CLC
ROR   R20
ROR   R20
```

(b)
```
LDI   R20, 0x39
SEC
ROL   R20
ROL   R0
```

(c)
```
CLC
LDI   R20, 0x4D
SWAP  R20
ROL   R20
ASR   R20
```

(d)
```
CP    R20, R20
LDI   R20, 0x7A
ROR   R20
```

25. Show the code to replace the SWAP instruction:
    (a) using the ROL instruction
    (b) using the ROR instruction
26. Write a program that finds the number of zeros in an 8-bit data item.

27. Write a program that finds the position of the first high in an 8-bit data item. The data is scanned from D0 to D7. Give the result for 68H.
28. Write a program that finds the position of the first high in an 8-bit data item. The data is scanned from D7 to D0. Give the result for 68H.

5.